25901182

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEBCA201 - INTRODUCTION TO MACHINE LEARNING
(2024 ADMISSION ONWARDS)
Scheme of valuation

Duration: 1 Hour **Maximum Marks: 35**

Part A Very Short Answer Questions. Answer any 5 questions.Each question carries 3 marks

1.. Distinguish between the four outcomes represented in a confusion matrix.

TP:Model correctly predicts positive class, TN: Model correctly predicts negative class, FP: Model predicted positive which is actually negative, FN: Model predicted negative which is actually positive. [Award 0.75 marks for each correct definition]

2.. Define centroid in K-Means clustering.

Concept of kmeans clustering ,value of k and its importance and intial selection of centroid – 2 marks

Concept of recalculation of centroid after each assignment of points – 1 mark

3.Show why ML is important today.

Automatic decision making / large dataset handling / any real world application – 3 marks

4. Explain the difference between hard margin and soft margin SVM.

Hard margin: No misclassification allowed but data must be perfectly separable. 1 mark

Soft margin: Allows some misclassification for better generalization, Soft margin works better with noisy datasets. 2 marks

5 Describe logistic regression.

Identify Logistic regression as supervised learning classification algorithm 2 marks

Output specifications as categorical/binary I mark

6 Compare clustering and classification.

Classification definition and should contain labelled dataset 1.5 marks

Clusteing --- unlabelled dataset 1.5 marks

7.Identify whether a given dataset is linearly separable or not.

If student correctly states whether data can/cannot be separated using a straight line, award marks. (Graphical illustration may also be accepted) 3 marks

Part B Short Answer Questions. Answer any 4 questions. Each question carries 5 marks

8. Analyze how multiple factors improve prediction accuracy in multiple linear regression.

Explanation should contain keywords like multiple independent variables, complex relationship, reduces prediction errors . Also award marks if explained with proper examples like house price prediction using multiple variables like area, location, rooms, etc. 5 marks

9.Explain the challenges faced while implementing ML models in real-world scenarios.

Availabily of data , data quality ,noise issue 2 mark

Computational cost / training time / lack of labeled data 1 mark

Overfitting/Underfitting issues 1 mark

Any other issues like security/ethical/deployment 1 mark

10. Build a Perceptron model for AND gate and draw the decision boundary.
Explanation of the model/equations/ weight,bias & activation function /AND gate truth table ---3 mark
Perceptron model diagram I mark
Proper diagram with decision boundary 1 marks

11. Explain the role of Machine Learning in image and speech recognition.
Explanation including importance and applications . 2 marks
Learn patterns, identify objects and speech and make intelligent decisions automatically. 3 marks
12. Write the Naive Bayes classifier algorithm and its working.
Definition 1 mark
Equation with explanation of terms 2 marks
Working and importance 2 marks

13. Identify the learning task: Disease positive or negative prediction. Justify.

Identification of the problem as classification task and the reasons 2 marks
Concepts of disease positive/negative prediction which is a supervised classification task since the output belongs to predefined classes (positive or negative) and the model learns using labeled training data. 3 marks